

RYOTA MAEDA

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github.com/elerac

Research Interests

- Computer Vision
- Polarimetric Imaging
- Computer Graphics
- Light Transport Analysis
- Computational Imaging
- 3D Reconstruction

Work Experience

Pohang University of Science and Technology (POSTECH)

Apr. 2026 – Present

Postdoctoral Researcher

Education

University of Hyogo

Apr. 2022 – Mar. 2026

Ph.D. of Engineering

Supervisor: Prof. Shinsaku Hiura

University of Hyogo

Apr. 2020 – Mar. 2022

Master of Engineering

Supervisor: Prof. Shinsaku Hiura

University of Hyogo

Apr. 2016 – Mar. 2020

Bachelor of Engineering

Publications

Broadband Hyperspectral 3D Imaging using Dispersed Structured Light

Jul. 2026

Suhyun Shin, Yunseong Moon, **Ryota Maeda**, David Lindell, Kyros Kutulakos, Seung-Hwan Baek
SIGGRAPH 2026

End-to-End Optimization of Polarimetric Measurement and Material Classifier

Mar. 2026

Ryota Maeda*, Naoki Arikawa*, Yutaka No, Shinsaku Hiura (**Equal contribution*)

21st International Conference on Computer Vision Theory and Applications (VISAPP 2026)

Hyperspectral Polarimetric BRDFs of Real-world Materials

Dec. 2025

Yunseong Moon, **Ryota Maeda**, Suhyun Shin, Inseung Hwang, Youngchan Kim, Min H. Kim, Seung-Hwan Baek
SIGGRAPH Asia 2025

Event Ellipsometer: Event-based Mueller-Matrix Video Imaging

Jun. 2025

Ryota Maeda, Yunseong Moon, Seung-Hwan Baek

IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR), *Highlight*

Dense Dispersed Structured Light for Hyperspectral 3D Imaging of Dynamic Scenes

Jun. 2025

Suhyun Shin, Seungwoo Yoon, **Ryota Maeda**, Seung-Hwan Baek

IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR)

Polarimetric Light Transport Analysis for Specular Inter-reflection

May. 2024

Ryota Maeda, Shinsaku Hiura

IEEE Transactions on Computational Imaging, 2024

Refinement of Hair Geometry by Strand Integration

Oct. 2023

Ryota Maeda, Kenshi Takayama, Takafumi Taketomi

Computer Graphics Forum (Proceedings of Pacific Graphics 2023)

EpiScope: Optical Separation of Reflected Components by Rotation of Polygonal Mirror

Dec. 2021

Ryota Maeda, Shinsaku Hiura

SIGGRAPH Asia 2021 Technical Communications

Visiting Research Experience

POSTECH Computational Imaging Group

Mar. 2024 – Feb. 2025

Visiting Research

11 months

Mentor: Prof. Seung-Hwan Baek

CyberAgent AI Lab

Aug. 2022 – Sep. 2022

Research Intern

2 months

Mentors: Dr. Kenshi Takayama and Dr. Takafumi Taketomi

Software on GitHub

Polanalyser | ☆ 210 stars

Polarization image analysis tool. Demosaicing, Stokes vector, Mueller matrix.

structuredlight | ☆ 166 stars

Generate and Decode structured light. Binary, Gray, XOR, Ramp, Phase-Shifting, Stripe.

EasyPySpin | ☆ 112 stars

cv2.VideoCapture like wrapper for FLIR Spinnaker SDK.

Skills

Programming: Python, C++

Hardware Development: Electronic circuit design, Embedded systems, 3D CAD

Visual Content Creation: Photography, Image processing, Scientific illustration

Language: Japanese (native), English (advanced)